

Statistical Analysis of the Disorder of Two-Dimensional Cellular Arrays in Directional Solidification

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By introducing a weighted Wigner-Seitz construction and, for the first time in directional solidification, using the minimal spanning tree (MST) approach and the (m, σ) -diagram, the statistical analysis of the topological defects and disorder of two-dimensional (2-D) cellular arrays has been carried out. For "standard" growth of massive Pb-30 wt pct Tl alloys, the underlying honeycomb has been brought out, which was rather unexpected, as the percentage of defects is so high that the cellular arrays are "melted" by the defects, with a structure close to that of a 2-D liquid. Furthermore, disorder can be described by a Gaussian noise applied on an array of hexagonal cells, Lewis's law is satisfied, and for a sufficiently large number of cells, a peak is evidenced in the primary spacing distribution, which indicates that, in some way, the cell size is selected. Nevertheless, the associated standard deviation is rather large. It is conceivable that the defects can have a leading role in the selection of the primary spacing.

I. INTRODUCTION

It has long been recognized that the physical properties of a material are strongly influenced by the microstructure which is formed in the solid phase during processing. Consequently, controlled solidification techniques that permit a close control of the microstructure have been extensively developed to produce engineering components. Therefore, it follows that the prediction of the microstructure as a function of the processing conditions is crucial in engineering. Besides, recent theoretical advances have greatly improved the current knowledge of the stability and selection of nonlinear structures in several pattern-forming instabilities observed in different areas of science^[1,2,3] that were previously disconnected. In directional solidification, the morphological instability of the planar solidification front^[4] results in the formation of cells and dendrites at the solid-liquid interface, which belong to this continuously expanding family. Apart from their outstanding role in practical applications, these patterns have thus received further attention on a more physical standpoint, as archetypes for the study of the nonlinear aspects of morphogenesis.

When a binary alloy, whose solute concentration is C_* , is solidified in an imposed temperature gradient G , a flat solid-liquid interface is no longer morphologically stable when the growth velocity V is above a critical value V_c . First, the solidification front develops a cellular microstructure.^[5] These cells advance parallel to the thermal gradient and, when a bulk sample is grown, self-organize into an array of columns (Figure 1(a)) which, by chemical etching on a transverse section, shows up as a two-dimensional (2-D) array (Figure 1(b)). For usual processing conditions, such arrays are never perfect and always contain a certain amount of topological defects.^[6]

In comparison with a few structures observed in other pattern-forming instabilities, *e.g.*, Rayleigh-Bénard con-

vection,^[7] Bénard-Marangoni convection, and nematic-shear-flow instability,^[8] the analysis of the organization of the cellular arrays is just beginning. On lead-30 wt pct thallium alloys grown upward by the Bridgman method, we have recently initiated such an analysis.^[6] By using Wigner-Seitz construction,^[9] the basic defects (polygons with less, or more, than six sides) have been identified and numbered. The major result was that the percentage of defects is high enough to induce a defect-mediated melting of the regular honeycomb, which is the "2-D solid" array predicted by nonlinear theories. Concomitantly, only a short-range order is detected by the correlation functions.

The main objective of the present article is the statistical analysis of the defects and disorder of the 2-D cellular arrays observed in directional solidification of a binary alloy. Indeed, the selection of the wave number in dissipative structures which are formed in far-from-equilibrium systems may be driven by disorder,^[10] which, moreover, may induce turbulence and chaos.^[11] The tools we have used to carry out this statistical analysis are described in Section II. The Wigner-Seitz construction is first improved by introducing a weighted construction. Then, the minimal spanning tree (MST) approach,^[12] which will provide complementary information, is explained. The array disorder is analyzed in Section III, and the basic question of primary spacing selection is discussed in Section IV.

II. TOOLS FOR THE STATISTICAL ANALYSIS OF DISORDER IN 2-D CELLULAR ARRAYS

A. Wigner-Seitz Construction

In order to be able to carry out the statistical analysis of the disorder of the cellular arrays which are observed in the directional solidification of a massive binary alloy, it is naturally indispensable to get either a picture of a good quality etching on a transverse section for the case of a metallic system (Figure 1(b)) or direct photographic or video recording if a transparent alloy is grown. Then, the array is acquired in 256 gray levels on a computer

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Manuscript submitted December 27, 1990.

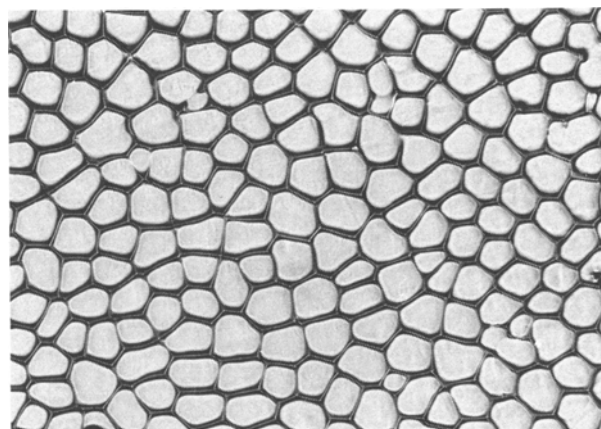
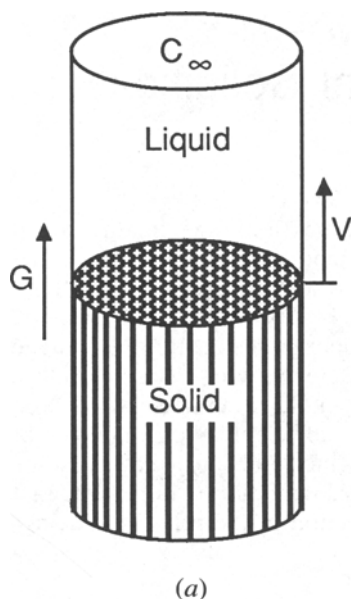


Fig. 1—(a) Schematic representation of cellular upward solidification of a cylindrical sample: the cells form an array of columns parallel to the growth direction. (b) Chemical etching on a transverse section. Pb-30 pct Tl. $G = 45^\circ\text{C}/\text{cm}$, $V = 1.1\text{ cm/h}$, and image width = 3.2 mm.

hard disk by means of a charge coupled device camera and a video digitizer. Appropriate filterings and scalings finally provide a binary image (Figure 2).

The black lines are superimposed on the cell walls. For the alloys we have investigated, their thickness, which merely results from the scaling procedure and is in no way related to the width of the intercellular grooves, never has appreciably influenced the subsequent array reconstructions. If it were the case, it would always be possible to reduce this thickness down to 1 pixel on the monitor. The cell centers, on which the Wigner-Seitz construction is based, are assimilated to the centers of gravity of the white areas in Figure 2.

The Wigner-Seitz construction is very popular under various denominations, among which Voronoi construction is certainly the most common.^[9] Nevertheless, the Wigner-Seitz construction has been preferred as it leads to Wigner-Seitz cells, whereas the result of a Voronoi construction are Voronoi polygons. Knowing the posi-

tions of the cell centers, the cellular array is constructed cell by cell by assigning to each center all points which are nearest to it. This is achieved by drawing the mid-perpendiculars to the segments joining any two centers (Figure 3). For each center, a "reconstructed" cell is defined as the smallest convex polygon surrounding it, which contains all points in space closest to that center. It has been proven that the construction is unique and fills space.^[13]

As they are truncated, all the cells that touch a side of the sampling window are discarded for the statistical analysis. Then, one can easily identify, classify, and number the defects; *i.e.*, the cells whose number of sides is different from six (Figure 4(a)). The percentage of defects d_i is defined by the ratio of the number of defective cells to the total number of cells.

Although it is a powerful tool, it is often inconvenient that the Wigner-Seitz construction has an intrinsic tendency to drive the array toward the average, which directly stems from the use of the segment mid-perpendiculars in the construction. This effect is not

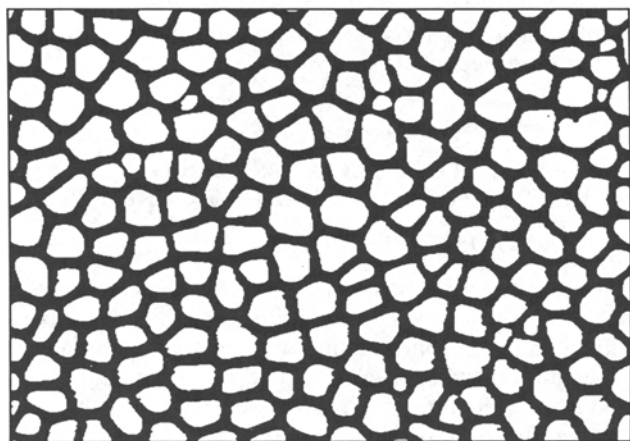


Fig. 2—Binary image of Fig. 1(b). Pb-30 pct Tl. $G = 45^\circ\text{C}/\text{cm}$, $V = 1.1\text{ cm/h}$, and image width = 3.2 mm.

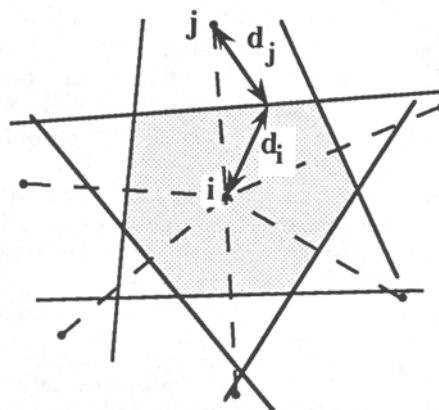
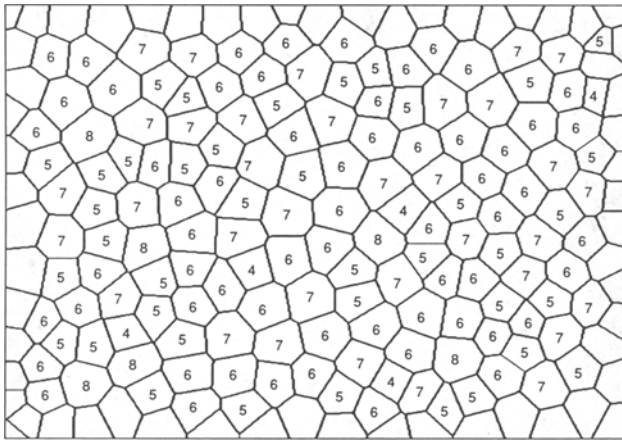
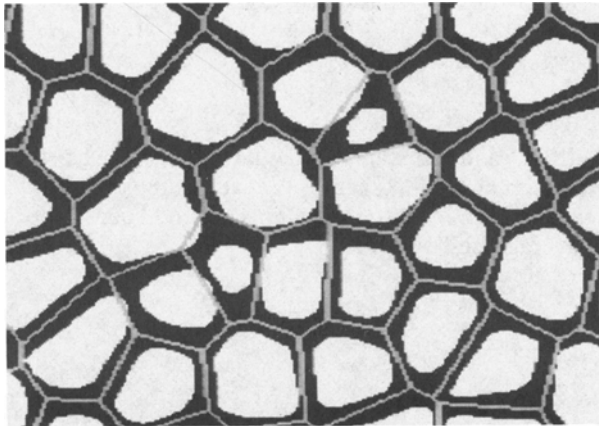


Fig. 3—Wigner-Seitz construction ($d_i = d_j$).



(a)



(b)

Fig. 4—Pb-30 pct Tl. $G = 45\text{ }^{\circ}\text{C/cm}$, and $V = 1.1\text{ cm/h}$. (a) 2-D cellular array obtained by the classical Wigner-Seitz construction. For each cell which is in full in the sampling window, the number of sides is given in the center. Image width = 3.2 mm. (b) Superposition of the 2-D cellular array obtained by the Wigner-Seitz construction on the binary image.

visible when the cells are of about the same size and the array fits well on the binary image, but the superposition is no longer satisfactory when a large cell and a small cell are neighbors, the latter then being enlarged at the expense of the former (Figure 4(b)). These findings led us to seek for some way of improving the classical Wigner-Seitz construction.

B. Weighted Wigner-Seitz Construction

From the mere observation of the original photograph (Figure 1(b)), the two characteristic parameters of each cell that are seen at a glance are its location and its size. Besides the coordinates of the center of gravity, it is thus rather natural to weight each cell by some power of the corresponding white surface on the binary image. Different attempts have been made with various powers, and the inverse of the square root of that surface turned out to be the relevant weight. It follows that the cell sides should be drawn one point after the other (Figure 5) by using the relation

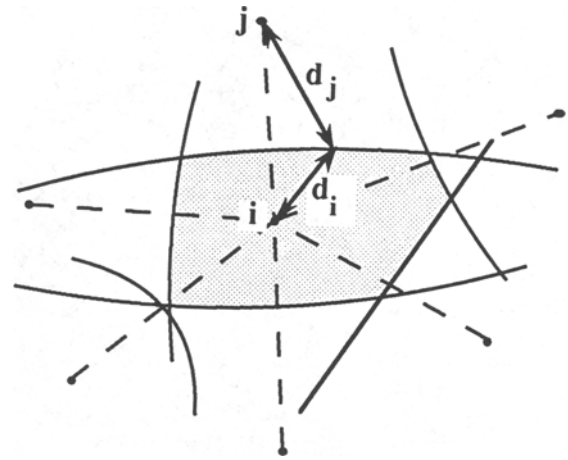


Fig. 5—Weighted Wigner-Seitz construction ($d_i/S_i^{1/2} = d_j/S_j^{1/2}$).

$$\frac{d_i}{d_j} = \sqrt{\frac{S_i}{S_j}} \quad [1]$$

where $d_i(d_j)$ is the distance from the point under consideration to the center of cell $i(j)$, whose surface is $S_i(S_j)$. In order to make clear the physical meaning of the weight that gives the best fit, it is convenient to replace each cell by a soft disk of identical surface. Indeed, as

$$\frac{d_i}{d_j} = \sqrt{\frac{S_i}{S_j}} = \frac{\phi_i}{\phi_j} \quad [2]$$

where $\phi_i(\phi_j)$ is the diameter of cell $i(j)$, it appears that one way to get the cellular array would be to compact a set of soft disks, just enough to suppress the interstices (Figure 6).

The 2-D cellular array is again constructed cell after cell (Figure 7(a)). Its superposition on the binary image is now excellent, even in places where the cell size locally changes very much (Figure 7(b)).

C. Minimal Spanning Tree

The graph theory is now well developed and is every day applied in a great variety of fields so that the basic definitions can be easily found elsewhere.^[12,14-16] Therefore, the following presentation is reduced to a minimum. Let us recall that an edge-weighted linear graph is composed of a set of nodes (the cell centers for the case of cellular solidification) and a set of edges, an edge being defined by a pair of cell centers, with a weight

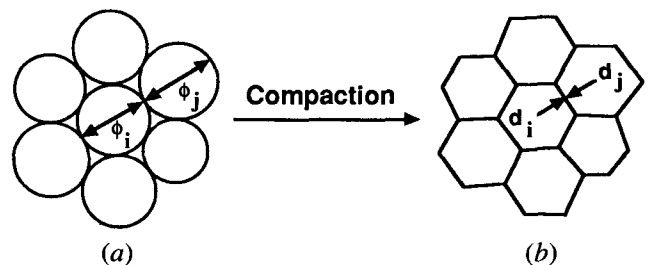
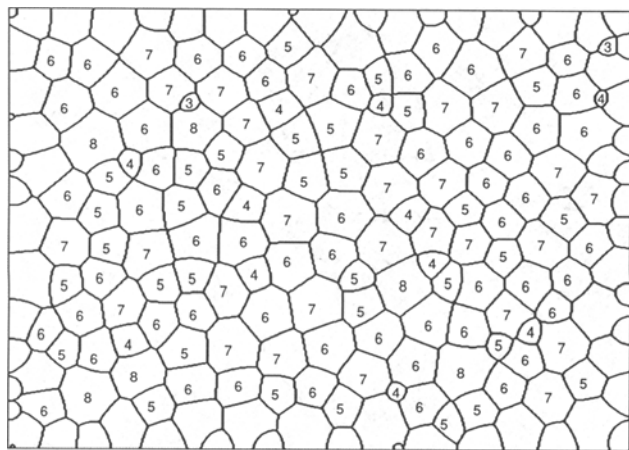
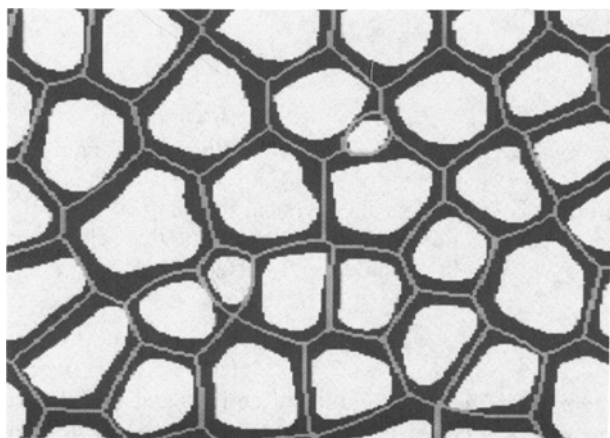


Fig. 6—(a) Compaction of a packing of soft disks to obtain a 2-D cellular array (b).



(a)



(b)

Fig. 7—Pb-30 pct Tl. $G = 45^\circ\text{C}/\text{cm}$, and $V = 1.1\text{ cm/h}$. (a) 2-D cellular array obtained by the weighted Wigner-Seitz construction. For each complete cell, the number of sides is given in the center. Image width = 3.2 mm. (b) Superposition of the 2-D cellular array obtained by the weighted Wigner-Seitz construction on the binary image.

assigned to each edge (the corresponding center-to-center distance). A MST is a connected graph without any closed loop, which contains all the cell centers and for which the sum of the edge weights is minimum. The most illustrative way to construct a MST is very pedestrian and thus leads to the most time-consuming algorithm. One starts from any cell center and adds at each step the cell center which is closest to the current tree, together with the weight associated to that shortest distance (Figure 8).

A MST can be constructed for any distribution of points on a surface. There are a few specific cases in which, for a given distribution of cell centers, the MST may not be unique and locally varies depending on the point which is selected to start its construction. Nevertheless, the important property is that all the possible MSTs are equivalent so that the edge-length histogram is unique, which legitimates the utilization of parameters that are deduced from the statistical analysis of that histogram to characterize the arrangement of the cell centers. The most informative parameters are linked to the moments of the distribution, the two major ones being the average edge

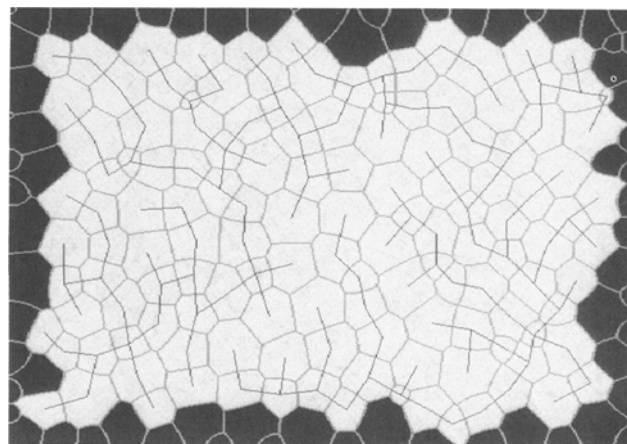


Fig. 8—The MST (in black) based on the cells in the white area is superimposed on the 2-D array obtained by the weighted Wigner-Seitz construction (the gray lines correspond to the cellular walls). Pb-30 pct Tl. $G = 45^\circ\text{C}/\text{cm}$, $V = 1.1\text{ cm/h}$, and image width = 3.2 mm.

length m^* and the standard deviation σ^* . It was shown by Dussert and co-workers^[12,16] that it is most convenient to normalize m^* and σ^* , roughly by the square root of the average cell surface $\langle S \rangle$,

$$m = \frac{m^*}{\sqrt{\langle S \rangle}} \frac{N-1}{N} \quad [3a]$$

$$\sigma = \frac{\sigma^*}{\sqrt{\langle S \rangle}} \frac{N-1}{N} \quad [3b]$$

and that the histogram of the dimensionless edge lengths m should rather be considered (Figure 9). By doing this, any distribution can be plotted in the (m, σ) -plane, where it can be compared to any other 2-D arrangement. For a meaningful comparison, one must be sure that converged values have been obtained for m and σ . Indeed, it will be seen in Section III-B that it is mandatory to determine these values with at least three significant figures for a good positioning in the (m, σ) -plane.

In order to check the convergence of m and σ , the total area S of the sampling window can be reduced by successive halvings and the MSTs constructed at each stage.

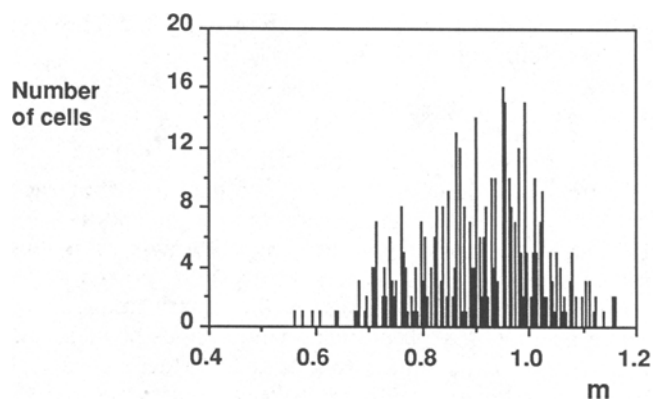


Fig. 9—Histogram of the normalized edge lengths m . Pb-30 pct Tl. $G = 45^\circ\text{C}/\text{cm}$, and $V = 0.75\text{ cm/h}$.

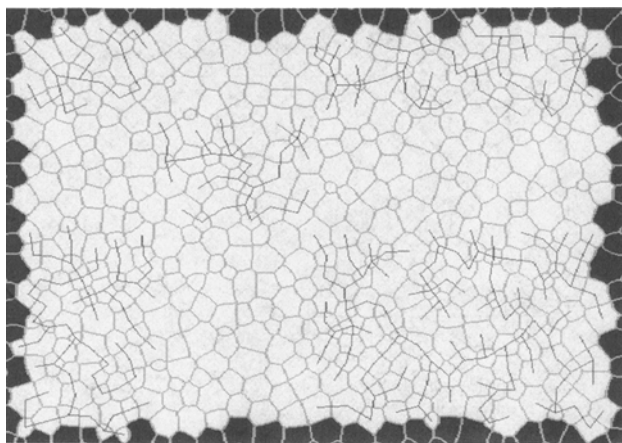
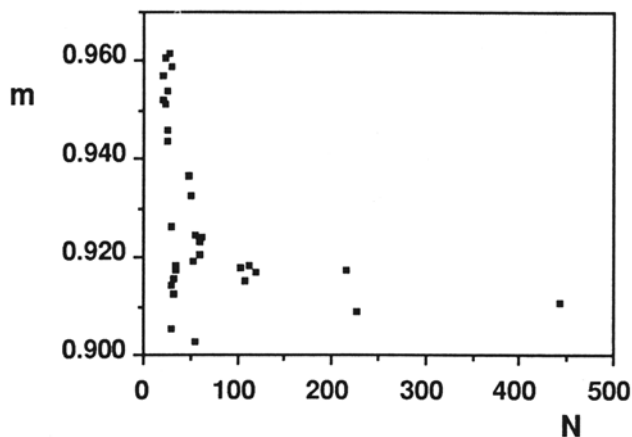


Fig. 10—Various MSTs (in black) constructed in limited sampling windows. Pb-30 pct Tl. $G = 45^\circ\text{C}/\text{cm}$, $V = 0.75\text{ cm/h}$, $N = 440$ cells, and image width = 3.3 mm.

For illustrative purposes, Figure 10 shows a combination of a few MSTs obtained at various steps. It is obvious that the smaller the sampling area, the stronger the influence of the too-long tree edges which are incorporated at the periphery. Indeed, as long as an area of finite lateral extent is considered, there is some overestimate of the value of m , which directly follows from the fact that the choice for the connection to the current tree is restricted to the cells which are in full in the sampling window. Proportionally, as the surface S diminishes, more natural edges are therefore impeded, and the departure from the real value of m increases.

Actually, the dispersion of the values is important for both m and σ when S is divided in 16 parts and continuously goes down as the sampling surface is enlarged (Figure 11). Nevertheless, the average values of these two parameters for a given division of S , $\langle m(S/2^n) \rangle$ and $\langle \sigma(S/2^n) \rangle$, have very different behaviors with $\langle N(S/2^n) \rangle$: the latter quantity is roughly constant, whereas the former slowly converges to its asymptotic value. For the lead-30 wt pct thallium alloy of Figure 10, a least-squares fit gives

$$\langle m \rangle = 0.908 + 0.461 \langle N \rangle^{-0.839} \quad [4]$$



with a correlation factor equal to unity. The asymptotic value of m is reached to a sufficient precision only for $S/2$, i.e., for a total number of cells $N = 200$ or thereabout. Consequently, the use of the (m, σ) -plane for a smaller number of cells would lead to erroneous conclusions on the disorder of 2-D cellular arrays.

III. ARRAY DISORDER IN CELLULAR SOLIDIFICATION OF Pb-30 WT PCT TI ALLOYS

A. Defects

As the 2-D cellular arrays constructed by the weighted Wigner-Seitz method are in better agreement with the binary image, this technique has been retained for the quantitative analysis.

The histogram of the n -sided cells (Figure 12) shows that the number of defects is extremely high, the six-sided cells being only 40 pct of all. The comparison with the values obtained by using the classical Wigner-Seitz construction reveals that the "liquid" nature of the cellular array is more pronounced with the weighted Wigner-Seitz construction (Table I), which reflects the correct elimination of the averaging effect. Besides, the percentage of the four-sided cells now is about 10 pct and that of the eight-sided cells about 5 pct. Previously, these percentages were of the order of 1 pct.^[6] When isolated in an otherwise perfect array of hexagonal cells, a pair formed by a five-sided cell and seven-sided cell is equivalent to an edge dislocation (Figure 13) so that it is worth associating the same gray level (or color) to complementary defects (Figure 14); i.e., the same gray level is given to n -sided cells and to $(12-n)$ -sided cells. Then, the liquid nature of the arrays is more easily visible: the darkest gray level, which is attributed to the six-sided cells, is not dominant and even does not assemble into grains slightly disoriented from one another and separated by grain boundaries constituted by chains of defect pairs, as it has been reported for other instabilities when the percentage of defects is intermediate.^[8] On the histogram of the center-to-center distances (Figure 15), a single peak appears, which means that the correlation between cells does not extend farther than the first neighbors.

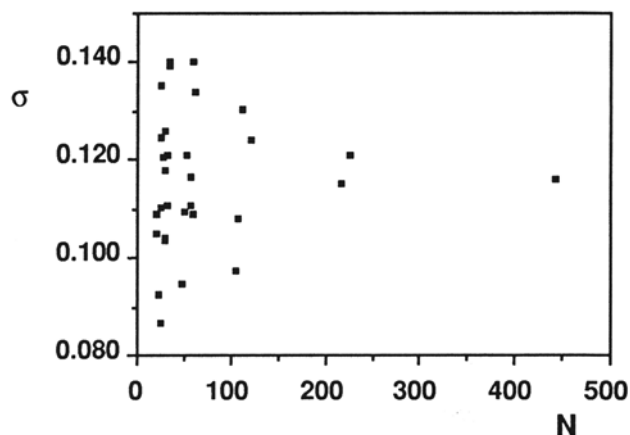


Fig. 11—Variation of m and σ with the total number of cells N . Pb-30 pct Tl. $G = 45^\circ\text{C}/\text{cm}$, and $V = 0.75\text{ cm/h}$.

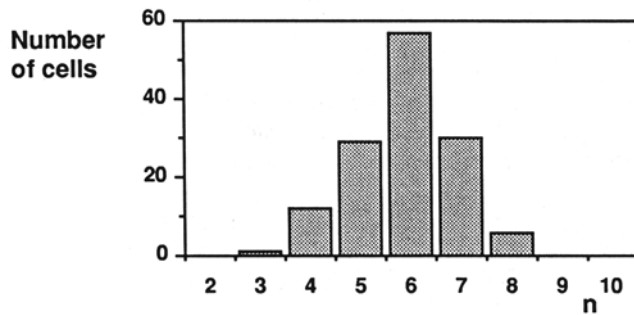


Fig. 12—Histogram of the n -sided cells. Pb-30 pct Tl. $G = 45^\circ\text{C}/\text{cm}$, and $V = 1.1\text{ cm/h}$.

The variation of the average surface of n -sided cells with the number of sides is given in Figure 16. It is most interesting that the average area $\langle S_n \rangle$ of an n -sided cell is a linear function of the number n of its sides which indicates that, in directional solidification too, the 2-D cellular arrays satisfy Lewis's rule. On the basis of observations on biological cells, this rule was first proposed by Lewis.^[17] Since, Rivier and Lissowski^[13] have demonstrated that Lewis's rule actually is a mathematical law which has to be satisfied by any space-filling structure. Furthermore, it has been shown that the slope $\mathcal{P}$ and intercept n_0 with the abscissa axis of Lewis's line are linked to a parameter $\mathcal{A}$, which measures the aging of the array,^[9] by the relations

$$\mathcal{P} = \frac{\partial \langle S_n \rangle}{\partial n} = \mathcal{A} \langle S \rangle \quad [5a]$$

$$n_0 - 6 + \frac{1}{\mathcal{A}} = 0 \quad [5b]$$

The intercept with the abscissa being about $n_0 = 2$, it stems from Eq. [5b] that $\mathcal{A}$ is close to 0.25, from which it can be concluded that the 2-D cellular arrays in that sense are relatively aged. This does not mean that there will be no further ripening of the array. Indeed, despite the fact that the aging parameter $\mathcal{A}$ nears the limit value and experiments show that the average cell surface $\langle S \rangle$ no longer changes, a continuous decrease of disorder, very likely by pair annihilation of defects, cannot yet be *a priori* excluded. Lewis's line would not be affected by such a process, in which the $\langle S_n \rangle$ are conserved.

The potentialities of Lewis's rule are not limited to the

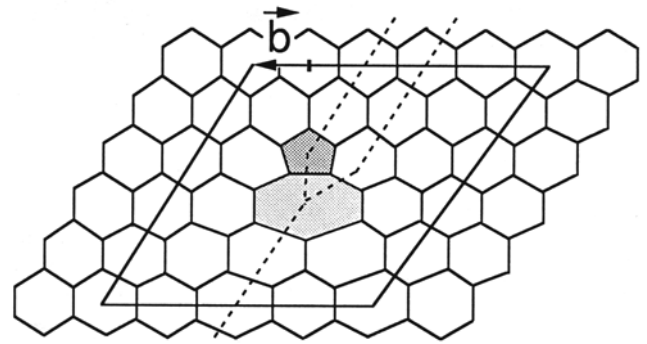


Fig. 13—Topological dislocation formed by a 5-7 pair of cells isolated in a honeycomb.

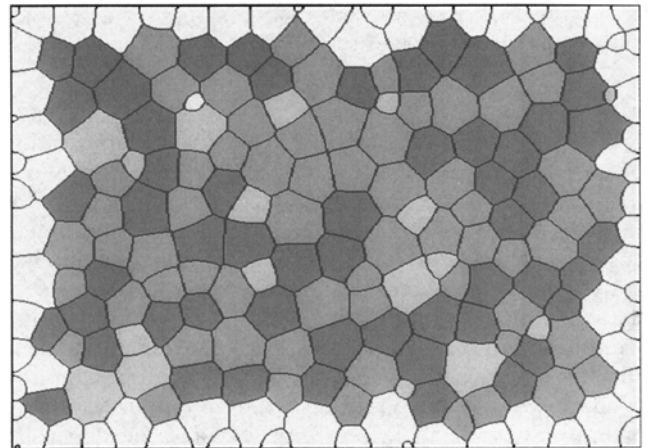


Fig. 14—Pb-30 pct Tl. $G = 45^\circ\text{C}/\text{cm}$, and $V = 1.1\text{ cm/h}$. 2-D cellular array obtained by the weighted Wigner-Seitz construction. The same gray level is attributed to complementary defects which distinctly show the "liquid" nature of the array as the darkest gray level, which corresponds to the six-sided cells, is not dominant. Image width = 3.2 mm.

above useful information. In addition, it is worth testing the quality of any reconstruction of a 2-D cellular array. For instance, in rapid solidification by laser scanning, the length scale of the cellular patterns is of the order of 100 nm so that the images are then obtained by electron microscopy, which reveals the intercellular precipitates. Most often, due to the local absence of precipitates, an appreciable number of cells are not completely closed so

Table I. Comparison of the Results of the Defect Analysis Based on the Weighted Wigner-Seitz Construction (wW-S) with Those Obtained by Using the Classical Wigner-Seitz Construction (W-S)

V (cm/h)	0.75		0.85		1.00		1.10	
	W-S	wW-S	W-S	wW-S	W-S	wW-S	W-S	wW-S
Pct defects	55	69	53	56	51	68	51	59
Pct three-sided cells	0	3	0	0	0	2	0	1
Pct four-sided cells	3	11	2	7	3	12	1	9
Pct five-sided cells	27	26	24	25	23	22	26	23
Pct six-sided cells	45	31	47	44	49	32	49	41
Pct seven-sided cells	22	19	25	21	22	24	22	22
Pct eight-sided cells	3	9	2	3	3	8	2	4
Pct nine-sided cells	0	1	0	0	0	0	0	0

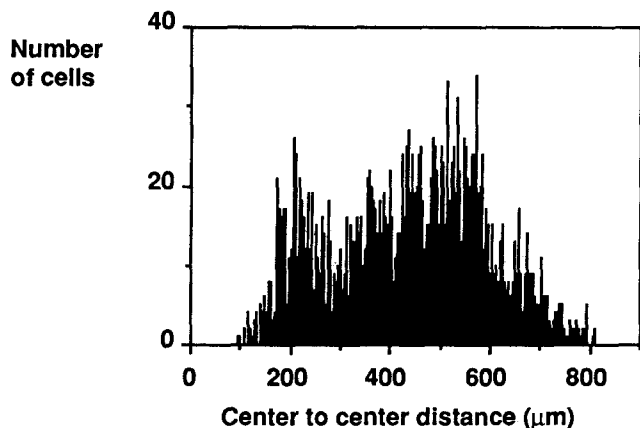


Fig. 15—Histogram of the center-to-center distances. Pb-30 pct Tl. $G = 45^\circ\text{C}/\text{cm}$, and $V = 1.1\text{ cm/h}$.

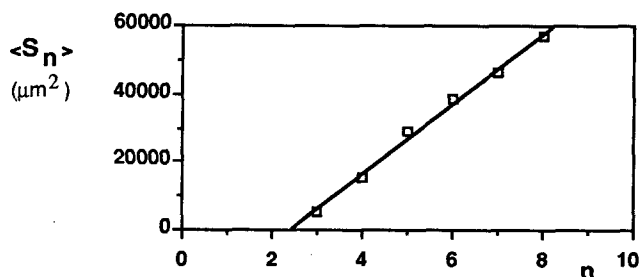


Fig. 16—Variation of the average area of an n -sided cell with the number n of its sides. Pb-30 pct Tl. $G = 45^\circ\text{C}/\text{cm}$, and $V = 1.1\text{ cm/h}$.

that the arrays which are directly constructed on these images do not satisfy Lewis's rule. It is only after the closure of these imperfectly delimited cells that this rule is fulfilled.^[18]

B. Order Behind Disorder

The 11 possible regular mosaics ($\sigma = 0$), quasi-periodic tilings, and the random distribution are plotted

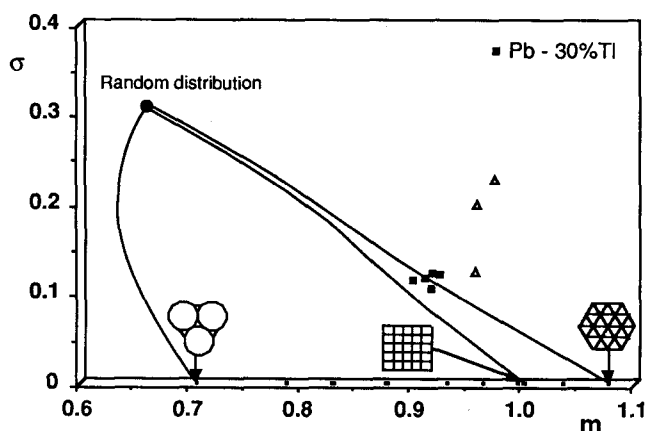


Fig. 17— (m, σ) -plane. The black squares on the horizontal axis correspond to the regular mosaics and the open triangles to a few quasi-periodic tilings.

in the (m, σ) -plane (Figure 17). Dussert and co-workers have progressively randomized regular lattices by giving each point a new position deduced from its previous one using a Gaussian distribution of increasing standard deviation.^[12,16] The computed trajectories joining some regular mosaics (triangular, square, etc.) to the random distribution are also shown in Figure 17. It follows that by constructing the MST, normalizing and plotting in the (m, σ) -plane, one is able to determine the underlying arrangement of the 2-D cellular arrays, *i.e.*, to elucidate order behind disorder.

It is remarkable that the points obtained for the Pb-30 pct Tl alloys precisely fall on such a trajectory which means that, despite the very high percentage of defects and liquid-type correlation functions, the underlying mosaic still manifests itself to a detectable extent. Figure 17 indicates that the cell centers basically form a triangular array, as expected for hexagonal cells, and that the disorder associated with the topological defects can be described by a Gaussian "noise," which might be a way to incorporate disorder in the theory.

Figure 18 shows the variation of m and σ with N . It is clear that the data points tend toward the line of an underlying triangular array as N is increased. When the total number of cells is very small (about 30) after division of the sampling window into 16 parts, there is an important risk of confusion with a perfect octagonal quasiperiodic tiling (open triangle) so that the necessity of using converged values for m and σ should be once more emphasized. Furthermore, Figure 18 gives us the opportunity to illustrate what in our opinion presently is the most severe limitation to the use of the (m, σ) -plane: the lack of any measure of the distances. Up to now, sound conclusions are permitted only when the experimental point coincides with some reference arrangement or trajectory. For instance, it is impossible to decide whether the octagonal tiling and the crosses that surround it are close to or far from the line. Work is presently in progress to introduce some metrics in the (m, σ) -plane.^[19]

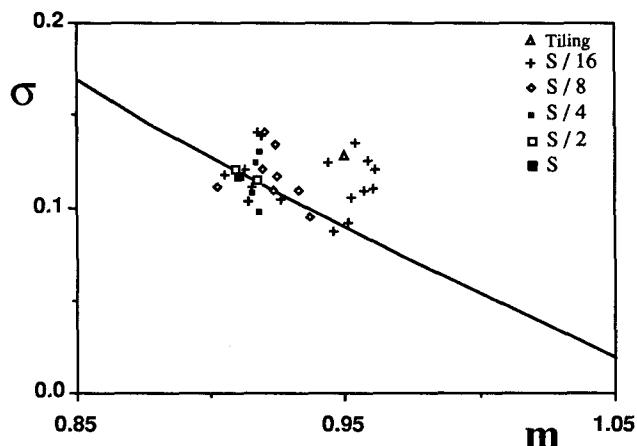


Fig. 18—Variation in the (m, σ) -plane of m and σ as the surface is reduced by successive halvings of the sampling window S . The open triangle corresponds to the quasiperiodic octagonal tiling. Pb-30 pct Tl. $G = 45^\circ\text{C}/\text{cm}$, and $V = 0.75\text{ cm/h}$.

IV. PRIMARY SPACING SELECTION

For bulk samples, the primary spacing λ of cellular and dendritic patterns is generally related to the nearest neighbor spacing, which is proportional to the square root of the average cell surface $\langle S \rangle$:^[20]

$$\lambda = \alpha \langle S \rangle^{1/2} \quad [6]$$

where α is a coefficient which depends on the arrangement. The determination of α is simple for a perfectly periodic array (e.g., $\alpha = 1.075$ for hexagons, 1 for squares), with long-range order, or for a random distribution ($\alpha = 0.5$) of cells. For patterns that are highly disordered without being completely random, the MST approach shows that the precise estimate of the coefficient α in Eq. [6] is no longer feasible. For practical applications, such as for the lead-thallium alloys which are under study, it is then better to consider that the value of the parameter α is unknown. In such a situation of intense disorder, it is far more grounded to characterize each cell by the diameter ϕ of the disk of equal area and to extract λ from the distribution of ϕ . The major advantage of this procedure is to circumvent the introduction of the parameter α , which is always more or less affected by subjectivity and, furthermore, makes difficult the comparison or merging of data from different origins.

For a set of 440 cells, the histogram of the diameter ϕ is given in Figure 19. To a good approximation, the distribution is symmetrical with a mean value $\langle \phi \rangle = 130 \mu\text{m}$, which is identified with the primary spacing λ . The standard deviation $\Delta\phi = 30 \mu\text{m}$ measures the dispersion $\Delta\lambda$ of the data. Although $\Delta\lambda/\lambda = 23 \text{ pct}$, it follows from the mere presence of a peak that, by some mechanism, the spacing is actually selected in these experiments. In the cellular range, the most advanced theoretical studies^[21] predict the existence of a band, whose width may be narrow, of allowable steady states rather than the sharp selection of a unique spacing. Nevertheless, to our knowledge, the question of the stability of these steady states is up to now not definitely answered. Recently, Warren and Langer^[22] have argued that the selected state depends on the sequence of events by which the system has been set into motion.

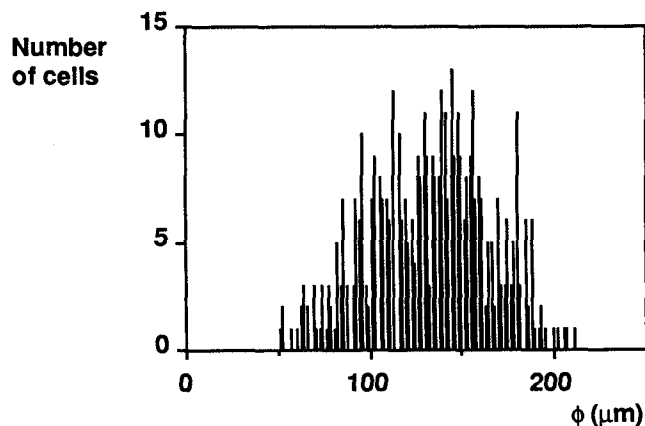


Fig. 19—Histogram of the diameter ϕ . Pb-30 pct Tl. $G = 45^\circ\text{C}/\text{cm}$, and $V = 0.75 \text{ cm/h}$.

It is of prime importance to realize that the shape of the distribution of ϕ strongly depends on the total number N of cells. For these lead-30 wt pct thallium alloys, a sound analysis of the selection of the primary spacing is possible only for $N \geq 200$. Indeed, there is a progressive disappearance of the peak as N is decreased. For N of the order of 200, the peak is still distinctly visible (Figure 20(a)). It becomes blurred about $N = 100$ (Figure 20(b)). The aspect of the distribution is extremely diverse when N is about 30, most often very flat (Figure 20(c)) and sometimes suggesting a peak (Figure 20(d)). It cannot be claimed that such an evolution of the histogram of the cell diameter ϕ with N is absolutely general but it is likely to be valid for many other binary systems.

V. CONCLUSIONS

A thorough analysis of the topological defects and disorder of 2-D cellular arrays has been carried out by introducing a weighted Wigner-Seitz construction and, for the first time in directional solidification, using the MST approach, which leads to the (m, σ) -diagram. For "standard" directional solidification of massive Pb-30 wt pct Tl alloys, in which the growth velocity V is suddenly switched on after thermal stabilization, the underlying honeycomb has been brought out, which was rather difficult to anticipate as the percentage of defects is so high that the cellular arrays are "melted" by the defects, with a structure close to that of a 2-D liquid. It might be important for theory that disorder can be described by a Gaussian noise applied to an array of hexagonal cells.

In experiments, interface microstructures which are steady on the average usually are achieved after a short period of growth, but the characteristic time τ for the array to reach the state of order that corresponds to the imposed value of the level of morphological instability $\nu (=V/V_c)$ is absolutely not known. If the time τ were shorter than the time necessary to reach a constant $\langle \phi \rangle$, this would mean that a high degree of disorder is intrinsic to the cellular arrays and that $\Delta\phi$ is always large. Such a scenario is in contradistinction with the present understanding of pattern-forming instabilities so that its likelihood is weak.

Indeed, a different behavior is, for instance, observed for Rayleigh-Bénard structures,^[7] Bénard-Marangoni and nematic-shear-flow cells.^[8] Just above the threshold, the patterns are well organized, and disorder is gradually enhanced by instability. It is furthermore well recognized that the percentage of defects is highly sensitive to the experimental procedure which is employed to establish the structures.^[7] As our results are for standard experiments, which probably are in a transient state of array organization, it is still an open question whether an optimized procedure is feasible or not. More explicitly, it would be desirable to know how the percentage of defects d , m , and σ vary with time. If there is some analogy with Bénard-Marangoni convection,^[23] it is likely that the transient elimination of defects, by pair annihilation and on the periphery, will induce a slow continuous decrease of d , down to a limiting value, which is the sole value characteristic of ν . Simultaneously, the (m, σ) -point will move toward the underlying mosaic.

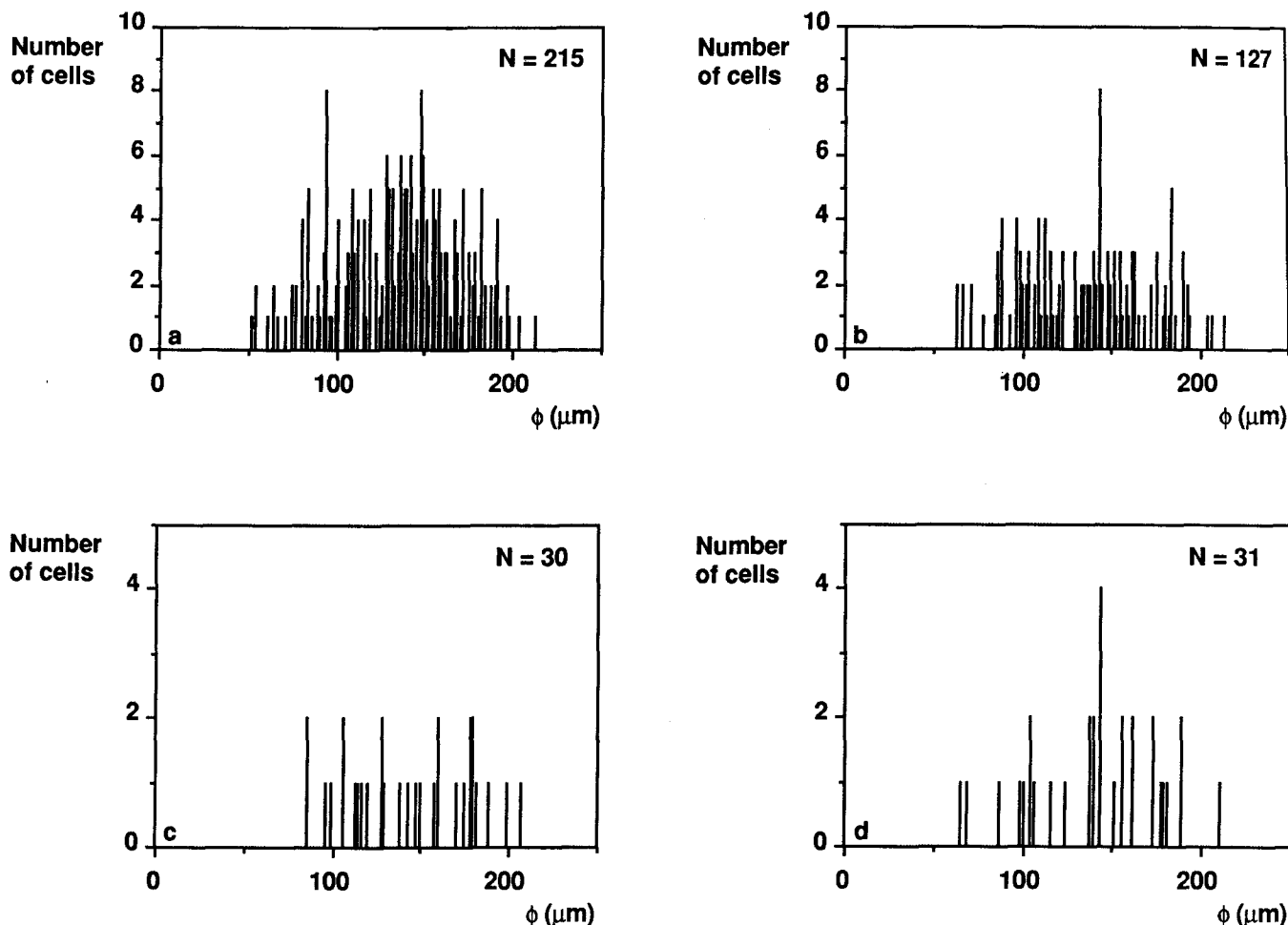


Fig. 20—Variation of the histogram of the diameter ϕ with the total number of cells N . Pb-30 pct Ti. $G = 45$ °C/cm, and $V = 0.75$ cm/h.

Then $\Delta\phi$ will be representative of ν , which will provide critical information on the sharpness of the selection of the primary spacing. The fact that the cellular arrays are far from being perfect honeycombs might be a clue to the understanding of the mechanisms of pattern selection.

Consequently, in order to know if the selection of a particular wavelength is defect mediated or not, it would be timely to clarify the role played by the defects in the dynamics of the adjustment of the average primary spacing to a steady-state value. The main experimental difficulty arises from the fact that, contrary to the Bénard-Marangoni instability in which the evolution of the array is visible at the top *surface*, the solid-liquid interface actually is a *front* in perpetual motion. It follows that the observations are necessarily limited to phenomena whose timescale is compatible with the finite length of the samples that are grown. Moreover, there always exists a certain latitude for displacements of the cell tips in the vertical direction so that the array dynamics are not contained in full in transverse sections. Work is currently in progress on metallic and transparent alloys. On the ground, some convection in the melt is almost unavoidable in the cellular range during the solidification at low rates of massive specimens^[24,25] so that microgravity experiments, in which the transport of heat and species is dominated by

diffusion, are under preparation for runs in a space shuttle to take place during the D2 flight in 1993.

ACKNOWLEDGMENTS

The support of the Centre National d'Etudes Spatiales and the Centre National de la Recherche Scientifique is gratefully acknowledged. The authors are moreover indebted to C. Dussert, W. Kurz, G. Rasigni, and R. Trivedi for many stimulating discussions.

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